يوسف السيد محمد عاشور ← سجل من ثانية

التاريخ: / /

الموضوع:

Q1: Add (101101) to (100111)

101101
+ 100111

1010100

Answer: 1010100

Q2: convert $(B65f)_{10}$ from hexadecimal to decimal

- B = 11, 6 = 6, 5 = 5, f = 15
- cal: $(11*16^3) + (6*16^2) + (5*16^1) + (15*16^0)$
- $= (11*4096) + (6*256) + (5*16) + (15*1)$

$= 45056 + 1536 + 80 + 15$

$= 46687_{10}$

Answer: 46687.

Q3: find 9's complement of $(546700)_{10}$

- 9's complement is calculated by ~~each~~ subtracting each digit from 9

number: 546700

9's complement $(999999 - 546700) = 453299$

التاريخ:

الموضوع:

Q4: simplify the Boolean function: $f = (x+y)(x+y')$
APPlY the Distributive law

$f = (x+y)(x+y')$
$= x + (y \cdot y')$
$= x + 0 \qquad (y \cdot y' = 0)$
$= x$

Answer x

Q5:
given: $f = AB + A'C$

$f = AB + A'C$
$f' = (AB + A'C)' = (AB)' \cdot (A'C)'$
$f' = (A' + B') \cdot (A + C')$
$f = [(A' + B') \cdot (A + C')]'$

Pos (maxterm) form: $(A' + B') \cdot (A + C')$

Q6:

Logisim: main of Untitled

File Edit Project Simulate Window Help

- Untitled*
  - main
  - Wiring
  - Gates
    - NOT Gate
    - Buffer
    - AND Gate
    - OR Gate
    - NAND Gate
    - NOR Gate
    - XOR Gate
    - XNOR Gate
    - Odd Parity
    - Even Parity
    - Controlled Buffer
    - Controlled Inverter
  - Plexers
  - Arithmetic
  - Memory
  - Input/Output
  - Base

Label

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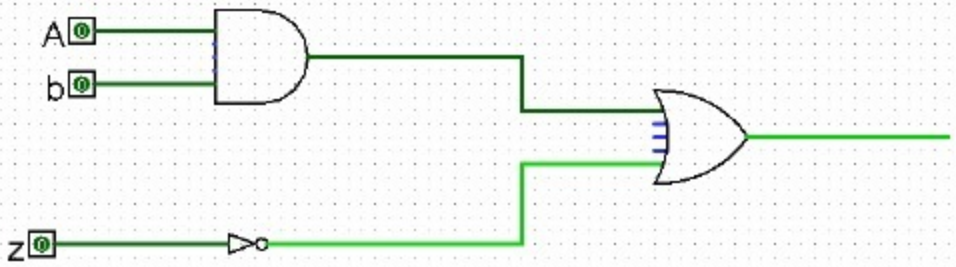
Horizontal Alignment	Center
Vertical Alignment	Base



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